Eligible Work as a Routing Variable: Measuring CPU/GPU Exhaustive Crossovers in Filtered Vector Search

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September 2026

Abstract

Filtered vector search is usually discussed as an index problem. In an embedded system, it is also a routing problem: after a predicate has reduced a collection to E eligible vectors, should exhaustive execution remain on the CPU or move to the GPU, and what does current evidence say about where ANN may enter that routing map? This paper measures the CPU/GPU boundary in Qenlo, a Rust/WGSL prototype whose canonical rows are independent of its search accelerators. The primary experiment holds data, queries, dimension, batch size, and hardware fixed. Across a post-hoc six-point sweep on an RTX 4050, shipped CPU and WGPU exhaustive search reverse order between $E = 2{,}000$ and 3,000 for 384-dimensional vectors. At $E = 100{,}000$, WGPU predicate evaluation reaches 3.240 ms P95 versus 16.657 ms on the CPU; at $E = 1{,}000$, the CPU reaches 0.230 ms while the same predicate path takes 2.883 ms. Within WGPU, compact rows beat shader-side evaluation at $E = 1{,}000$, then lose at $E = 100{,}000$, so routing also includes representation choice. At the dense endpoint, the bootstrap interval for the latency ratio between a tuned USearch HNSW point and exhaustive WGPU crosses parity; recall@10 is 0.99224 and 0.99998, respectively, against an FP64 oracle. Android and Intel Arc records reproduce the dense CPU/GPU ordering but do not identify a controlled mobile crossover. An exploratory A6000 study finds a separate FAISS-versus-CUDA-prototype reversal between 10,000 and 100,000 eligible 768-dimensional vectors; it is not a shipped-Qenlo result. The artifact contains 563,644 timed observations, of which 558,644 satisfy their stated recall qualification. The evidence supports a hardware- and implementation-specific phase map, not a universal claim that exhaustive search beats ANN.

1 Introduction

The practical input to a filtered nearest-neighbor query is not just collection size N. A metadata predicate first defines an eligible set S_P, and only $E = |S_P|$ vectors can enter the answer. Exhaustive scoring therefore performs $O(ED)$ arithmetic for dimension D. That makes a million-row collection with a one-percent filter a different execution problem from an unfiltered million-row query, even when both live in the same database.

Recent filtered-ANN work already shows that selectivity, filter/query correlation, and implementation choices can change the best plan [1, 4, 5, 9, 10]. The contribution here is narrower. This paper measures the completed-call boundary between CPU and portable-GPU exhaustive execution after filtering, including dispatch, synchronization, and bounded result readback. It also asks whether one tuned high-recall HNSW point displaces that boundary. The answer is not a new index. It is an empirical routing map with explicit gaps.

The study makes four contributions:

1. a controlled native CPU/WGPU sweep that brackets a winner reversal to $(2{,}000, 3{,}000)$ eligible vectors on one RTX 4050 host;
2. an ablation showing that eligible-row materialization, shader-side predicate evaluation, and mask transfer have different fixed costs, so E alone is not a sufficient routing feature;
3. external Android and Intel Arc observations plus a separately labeled A6000 CUDA/FAISS study, with no cross-cohort pooling; and
4. a retained evidence ledger containing raw calls, failures, checksums, tuning/evaluation separation, and an independent FP64 oracle where claimed.

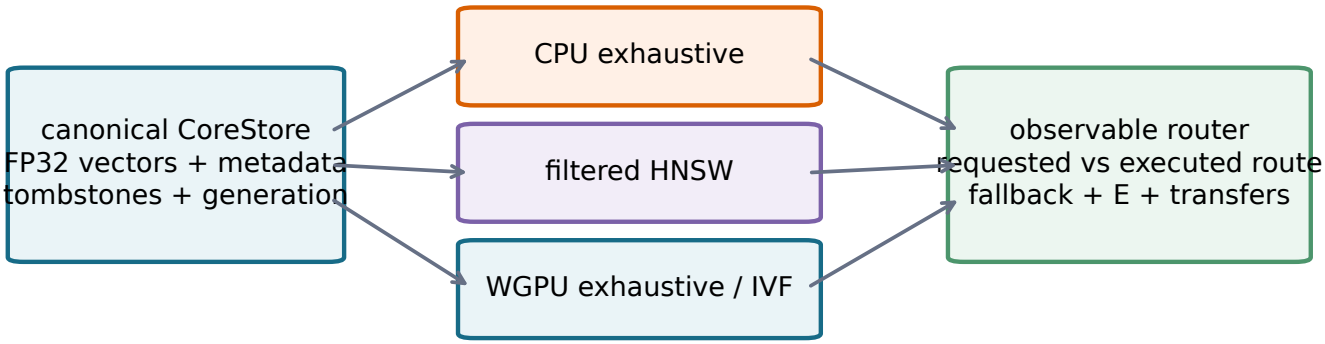

Figure 1: Canonical rows define logical truth. CPU, HNSW, and WGPU search structures are disposable accelerators behind an observable router.

The central research question is therefore: *for a fixed implementation, device, dimension, and batch policy, where does eligible work reverse the preferred exhaustive execution route, and how does predicate representation shift that boundary?* ANN is secondary: the retained points test whether approximate search can yet be placed on the same map.

2 Scope and terminology

Exhaustive versus numerically exact. The CPU and GPU paths score every eligible candidate. We call them *exhaustive*. Their floating-point order can differ from an FP64 oracle near ties, so the paper reports oracle recall instead of using "exact recall" as a numerical claim. The dense native cells achieve recall@10=0.99998; the sparse native cells and qualified A6000 cells achieve 1.0.

Completed-call latency. The timed boundary begins immediately before the search call and ends after ordered identifiers and distances are available on the host. It includes predicate work inside the selected route, query transfer, dispatch, synchronization, readback, and host merge. Index build, corpus loading, and oracle construction are outside timing. Kernel-event measurements are not compared with completed calls.

For the native compact-row sweep, the eligible list is neither precomputed nor reused across queries. Each timed batch-one call first traverses the predicate to count eligible rows, then the current WGPU path traverses it again to materialize a fresh row-ID list. The query and row IDs are uploaded into reusable scratch buffers for that call; only the vector chunks and allocations remain resident. The reported compact-row latency therefore includes both predicate traversals, compaction, ID and query upload, scoring, synchronization, readback, and host merge. It does not measure a cached-list route. This duplicated filtering is an implementation cost of the tested revision, not a requirement of compact-row execution.

Non-pooling. The native Windows, Android, Intel Arc, Linux library, real A6000, and synthetic A6000 records differ in hardware, data, implementation, or evidence quality. Every numerical comparison stays within a cohort. Cross-cohort agreement is treated as replication of a direction, not as a pooled effect.

3 System boundary

Qenlo's canonical `CoreStore` owns normalized FP32 vectors, metadata, public identifiers, tombstones, and the committed generation. Exact CPU state, filtered HNSW, and WGPU buffers are derived and rebuildable (Figure 1). This separation matters to the research question: route changes do not alter logical existence or recovery semantics.

The CPU path uses AVX2/FMA when available and accumulates dot products in FP64. The WGPU backend has three exhaustive predicate forms: upload a compact eligible-row list, upload a mask, or evaluate a supported predicate in WGSL. Vectors remain device-resident. The shader scans bounded chunks, each chunk emits at most k candidates, and the host merges those lists. If an item is outside its chunk top k, at least k items in that chunk

Table 1: Retained evidence used by the paper. "Aggregate" means raw per-call series were not supplied.

Cohort	Timed obs.	$N \times D$	Hardware	Evidence
Windows endpoints	225,000	100k×384	RTX 4050	raw calls
Windows crossover	300,000	100k×384	RTX 4050	raw calls
Linux libraries	10,500	100k×384	cloud CPU	raw calls
Real strict gate	20,000	1M×768	RTX A6000	raw calls
Synthetic held-out	4,000	1M×768	RTX A6000	raw calls
Android device lab	4,144	10k/100k×384	Mali-G615	aggregate
Total	563,644			

precede it under the common distance/ID order; the item cannot enter the global top k. This proves the chunk merge but is not claimed as algorithmic novelty.

The automatic route reports requested and executed backends, fallback, filter strategy, eligible/scored counts, transfer bytes, owned allocations, and timing phases. Required-GPU mode fails rather than silently falling back. These diagnostics let the benchmark distinguish an intended GPU run from a CPU fallback.

4 Experimental method

4.1 Evidence inventory

Table 1 summarizes the retained observations used in this revision. The Windows endpoint total includes nine completed configurations; the crossover campaign adds twelve CPU/WGPU cells. The 2,506-sample incomplete CPU development run and duplicated cloud mirrors are excluded. The real A6000 total includes a 5,000-sample cuVS run retained as a correctness-gate failure. Thus 558,644 of the 563,644 timed observations satisfy their stated recall qualification.

4.2 Primary native cohort

The Windows cohort uses 100,000 384-dimensional AG News sentence embeddings derived from a public DTU artifact [8]. Corpus, tuning, and evaluation query ranges are disjoint. Synthetic metadata is independent of vector geometry. Every retained configuration uses $k = 10$, batch one, 200 warmups, 5,000 held-out queries, and five repetitions. The endpoint cells use $E = 1{,}000$ and 100,000. The later localization sweep uses $E \in \{2k, 3k, 4k, 6k, 8k, 10k\}$ for CPU and compact-row WGPU. It was designed after inspecting the endpoints and is descriptive.

USearch expansion was selected on 1,000 tuning queries. $ef = 128$ reached tuning recall@10=0.9932 and was then frozen for the held-out range. The independent oracle scores every eligible vector in FP64 outside timing and rejects duplicate, deleted, unknown, or predicate-ineligible results.

For the primary ratios, the unit of resampling is a complete run, not an individual query. A seeded 10,000-draw bootstrap reports percentile intervals for the ratio of median run-level P95 values. Five runs produce a coarse interval; it does not model device, driver, thermal, or dataset variation.

4.3 External and exploratory cohorts

The Android schema-v1 records identify Android 16/API 36, a Mediatek MT6897 CPU, Mali-G615 MC6/Vulkan, and app version 0.1.0. Quick, full, and soak suites contain 16, 64, and 512 reported samples per cell. All 21 cells report recall@10=1.0, no fallback, and no failure. Power source is absent and thermal state is the uninterpreted string "0". The aggregates were transcribed without reconstructing unavailable raw distributions.

The A6000 real cohort uses one million 768-dimensional ModernBERT vectors from TopK Bench [12]. Its strict predicate yields $E = 10{,}026$. NumPy CPU, FAISS GPU Flat, and a PyTorch/CUDA research prototype each have 5,000 completed calls and an independent FP64 oracle. cuVS returned recall 0.9 and is disqualified. Predicate materialization is outside this cohort's latency boundary because every system receives the same prefiltered matrix.

The synthetic A6000 cohort uses normalized random FP32 vectors, 100 queries, 50 warmups, five repetitions, and a fresh held-out seed. It measures $E \in \{1k, 10k, 100k, 1M\}$. FAISS and the research CUDA prototype search the same matrix and have complete unordered top-10 agreement. This is cross-implementation agreement, not an independent oracle. Native WGPU produced no A6000 samples because the container exposed no usable NVIDIA Vulkan ICD.

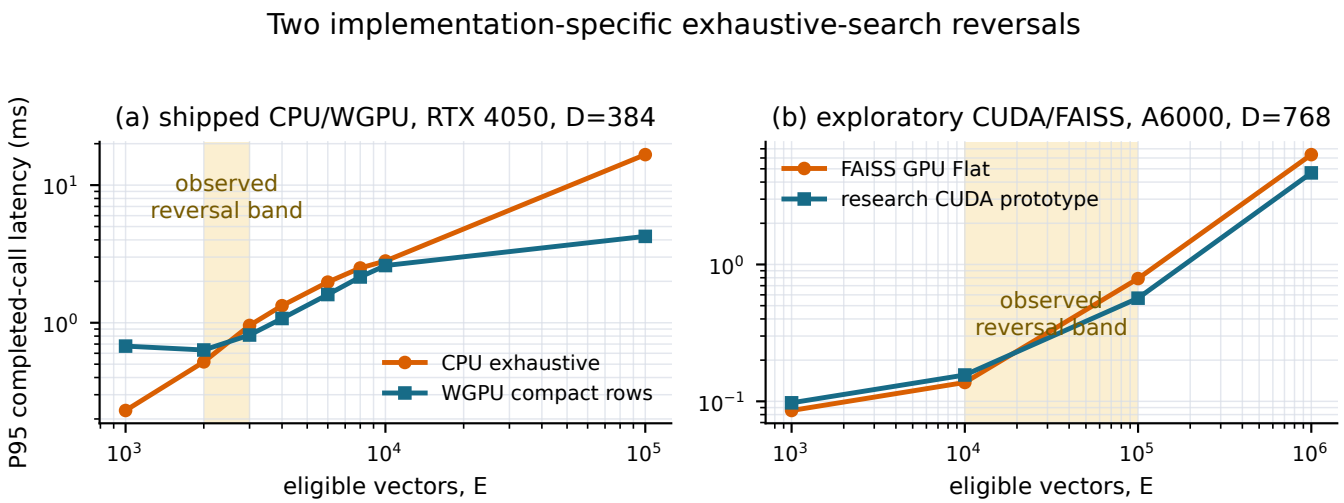

Figure 2: Two non-poolable implementation reversals. Lines connect measured points; shaded regions are brackets, not fitted universal thresholds. The left panel is shipped Rust/WGSL. The right panel is an exploratory PyTorch/CUDA prototype.

5 Results

5.1 A native exhaustive phase boundary

Figure 2 makes the two measured reversals literal while keeping them separate. On the RTX 4050, CPU wins at $E = 2{,}000$ (0.520 versus 0.634 ms P95), while WGPU compact rows wins at $E = 3{,}000$ (0.814 versus 0.956 ms). WGPU remains ahead through 10,000. The observed bracket is therefore $(2k, 3k)$ for this host, code revision, dimension, batch size, and predicate representation.

At $E = 100{,}000$, WGPU predicate search reaches 3.240 ms P95 versus 16.657 ms on CPU, a 5.14× ratio with whole-run bootstrap interval [3.55, 6.77]. At $E = 1{,}000$, CPU requires 0.230 ms, compact-row WGPU 0.677 ms, and shader-side predicate WGPU 2.883 ms. The endpoint reversal rules out a device-presence-only GPU routing policy. The WGPU win at $E = 3{,}000$ occurs despite the duplicated predicate traversal described in Section 2, so the measured bracket includes avoidable implementation overhead. It does not locate the crossover for a one-pass or cached-list path; that requires a matched ablation.

The six localization points do not support a transferable fitted crossover. A useful execution model is

$$T_{\text{CPU}} \approx \alpha_c + \beta_c ED, \tag{1}$$

$$T_{\text{GPU}} \approx \alpha_g(B) + \beta_g ED + T_{\text{materialize}}(P, E) + T_{\text{sync}}, \tag{2}$$

but the coefficients depend on route, memory behavior, and device state. Fitting them to six post-hoc P95 points would create false precision. The measured bracket is the stronger result.

5.2 Predicate representation is a routing variable

Figure 3 compares matched strategies. At $E = 1{,}000$, compact rows are 4.26× faster than shader-side predicate evaluation (0.677 versus 2.883 ms). At $E = 100{,}000$, the direction reverses: shader-side predicate evaluation is 1.31× faster than compact rows (3.240 versus 4.243 ms) and 1.38× faster than a mask (4.459 ms). Materializing a tiny eligible set can save device work; representing a dense set explicitly can cost more than evaluating the predicate in place. Route selection therefore has at least two decisions: choose CPU, GPU, or ANN, then, for GPU, choose rows, mask, or shader-side predicate evaluation. A router that uses only E misses this interaction.

At the dense endpoint, USearch reaches 3.625 ms P95 and recall@10=0.99224. WGPU predicate reaches 3.240 ms and 0.99998. The nominal latency ratio favors WGPU by 1.12×, but its whole-run bootstrap interval [0.77, 1.54] crosses parity. The result establishes one competitive high-recall operating point, not an ANN phase boundary. No recall–latency curve is available across E.

A separate Chroma HNSW cell on the same Windows data reaches 1.130 ms P95 and recall@10=0.99414. It uses Chroma's own storage and API rather than a route inside Qenlo, so it is not placed in Figure 3; it is retained in

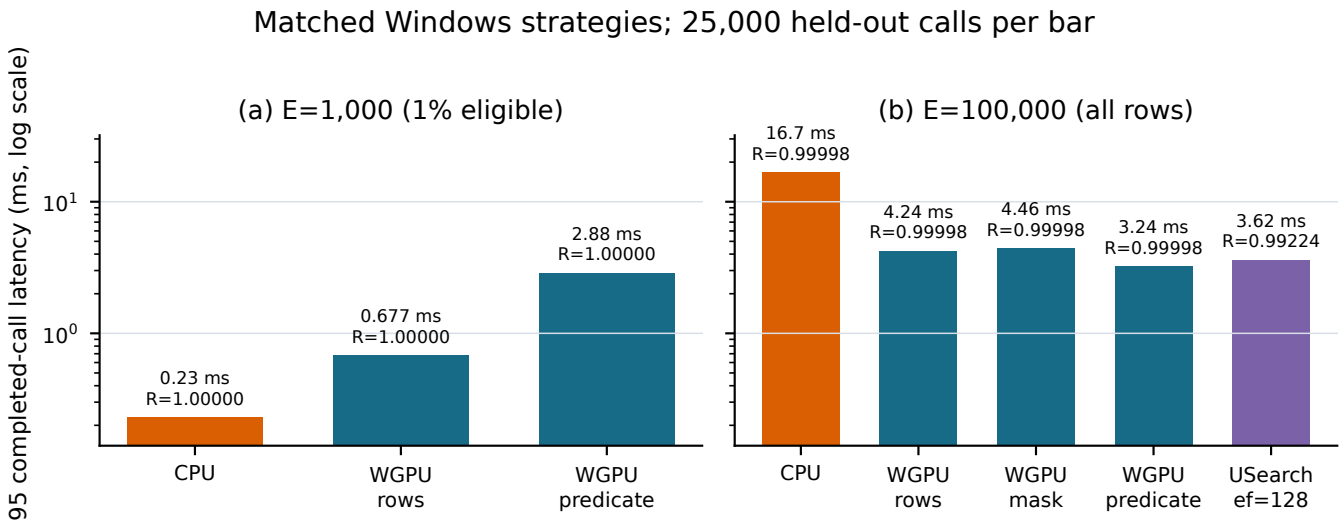

Figure 3: Matched native Windows strategies. All exhaustive results score the same eligible set; the USearch point is approximate. Recall is measured against the same FP64 oracle.

Appendix A. Its result matters: even at the dense endpoint where exhaustive WGPU is competitive with USearch, a different HNSW implementation is substantially faster. The evidence does not support "exhaustive beats ANN" as a system-independent statement.

5.3 Android and non-NVIDIA evidence

Figure 4 replaces the previous heterogeneous seven-bar chart with matched panels. CPU wins the 10k-row quick cell at P95 (5.767 versus 12.487 ms). WGPU wins the 100k full cell (16.975 versus 29.436 ms) and the 512-sample soak (17.678 versus 27.019 ms). Because both row count and suite length change, this is a qualitative reversal, not a mobile crossover bracket.

In the matched 100k soak, exhaustive WGPU also has lower P95 than IVF-Flat (21.453 ms) and IVF-SQ8 (37.090 ms). This does not prove that exhaustive search dominates IVF: only one parameterization and one generated workload were observed. Dense WGPU batch eight reports 11.131 ms versus 17.678 ms at batch one, but the emitted batch statistic is not normalized per query. It supports amortization as a route feature, not a per-query speedup claim.

An independent Intel Arc soak at 100k×384 reports 4.444 ms P95 for exhaustive WGPU and 16.486 ms for CPU, both at recall@10=1.0 over 512 samples. This is a second non-NVIDIA execution result. It is not enough to estimate a device-family effect.

5.4 The exploratory A6000 reversal

At the real TopK point $E = 10{,}026$, FAISS GPU Flat leads the qualified systems: 0.132 ms P95 versus 0.171 ms for the research CUDA exhaustive prototype and 0.197 ms for NumPy CPU. All three reach oracle recall@10=1.0 over 5,000 calls. The prototype is 1.29× slower than FAISS. Every manuscript occurrence now names it "research CUDA exhaustive prototype" because it is PyTorch code, not the shipped Qenlo backend.

The fresh-seed synthetic sweep produces a different result at larger E. FAISS is faster at 1k and 10k; the buffered prototype is faster at 100k (0.567 versus 0.789 ms) and 1M (4.668 versus 6.347 ms). The observed reversal lies in $(10k, 100k)$. The relative gain is stable at the two largest points (1.39× and 1.36×), but no intermediate cell localizes the boundary further. This is evidence about two GPU formulations, not a general PyTorch-versus-FAISS result.

5.5 What the broader archive adds

The descriptive Linux cohort holds data and host fixed for seven libraries, 1,500 calls each. FAISS HNSW is fastest at 1.494 ms P95 and recall 0.9994. Chroma HNSW reaches 3.777 ms/0.9984, USearch 5.093 ms/0.9914, FAISS Flat 10.217 ms/1.0, Qenlo CPU 12.901 ms/1.0, Milvus Lite AUTOINDEX 76.229 ms/0.9988, and LanceDB Flat 242.256 ms/1.0. Figure 5, moved to Appendix A, is context because build, persistence, and API semantics differ. The

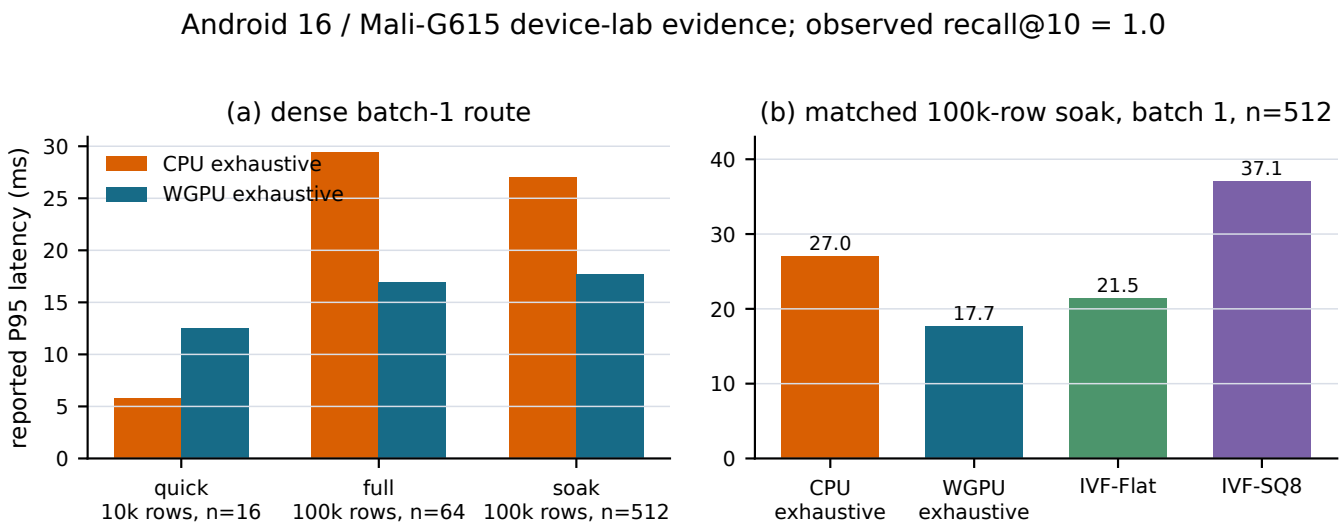

Figure 4: Android device-lab aggregates. Panel (a) reports dense batch-one CPU/WGPU cells. Panel (b) holds corpus, suite, and batch fixed. The IVF rows happened to return recall@10=1.0 on this workload; they remain approximate algorithms.

Table 2: Hypothesis status after the retained campaigns.

	Hypothesis	Status
H1	Exhaustive GPU can beat CPU and compete with high-recall HNSW	Partial: dense native point supports it; HNSW interval crosses parity
H2	E, not N, dominates exhaustive latency	Open: E varies, but matched E across N does not
H3	Fixed overhead creates a CPU/GPU reversal	Supported on RTX 4050; bracket is device-specific
H4	Batching shifts the reversal	Descriptive Android/Arc support; no matched native batch matrix
H5	High-recall ANN can lose its latency advantage	One controlled point only; no Pareto curve
H6	Adaptive routing reduces regret	Inconclusive: choices are plausible, counterfactual regret is unmeasured

cohort shows that implementation boundaries can dominate the label "exhaustive" or "ANN." It is not a product leaderboard.

6 Hypotheses and inference limits

The experiments were initially organized around six hypotheses. Table 2 states their final status without converting absent cells into positive evidence.

Three inferences survive this audit. First, eligible work is a better routing input than corpus size alone, but D, B, predicate representation, and device state remain necessary. Second, route selection must be calibrated to the concrete implementation. The native reversal occurs around a few thousand 384-dimensional vectors; the unrelated A6000 formulation reverses one to two orders of magnitude later at $D = 768$. Third, an ANN choice should be made from a recall–latency frontier, not one tuned point. The current paper cannot determine when filtered ANN becomes Pareto-optimal as E changes.

7 Related work and novelty boundary

FAISS provides optimized exhaustive and approximate CPU/GPU search [3, 6]. HNSW exposes a recall/latency tradeoff through graph expansion [7]. DiskANN and SPANN target much larger, storage-aware regimes [2, 11]. Filtered-DiskANN and ACORN modify graph construction or traversal to preserve access under predicates [4, 9].

Recent benchmarks make broad novelty claims about "exact sometimes wins" untenable. Iff et al. vary selectivity across modern FANNS methods and find that no method dominates every scenario [5]. Shi et al. emphasize unified tuning, query difficulty, selectivity, and dataset dependence [10]. Amanbayev et al. report hybrid approximate/exhaustive execution and pgvector cases where an optimizer selects ANN despite a comparable exhaustive plan with perfect recall [1].

Against that literature, this paper's novelty is the measured CPU-to-portable-GPU exhaustive boundary after predicate compaction, with completed-call costs and observable fallback, not the idea that selective filters can favor a scan. The present evidence is less complete than a FANNS benchmark: it lacks filter-aware ANN methods, recall curves across selectivity, correlation regimes, and scale. It is more specific about one embedded execution boundary and its artifact trail.

8 Threats to validity

The primary crossover uses one laptop, one dimension, one batch size, one corpus, and metadata independent of vector geometry. The six-point sweep is post-hoc. The compact-row path performs two predicate traversals per call in this revision, so removing the route-count pass or caching eligible rows could move the bracket. The CPU implementation uses AVX2/FMA with FP64 accumulation; it is a valid Qenlo baseline, not a state-of-the-art CPU boundary. FAISS CPU Flat, optimized FP32 SIMD, and multithreaded BLAS baselines are absent from the matched Windows sweep.

The Android evidence is author-supplied aggregate output without raw series, controlled thermal state, or power telemetry. The Intel Arc record is one device-lab soak. The A6000 prototype is unshipped and excludes predicate materialization in the real TopK cell. Process isolation between FAISS and PyTorch remains a threat despite matched data, query order, hardware, and timing boundary.

The study has one controlled HNSW operating point. It does not include a genuinely filter-aware ANN system in the native selectivity sweep. Metadata correlation, N at fixed E, $D = 768$ on the native host, concurrency, energy, memory peaks, and fixed-policy routing counterfactuals remain unmeasured. The correct next experiment is a same-host sweep over E, D, and recall that compares optimized CPU exhaustive, WGPU exhaustive, FAISS Flat, HNSW curves, and at least one filter-aware ANN method. Until then, the phase map has an ANN-shaped blank region.

9 Conclusion

The strongest result is small enough to state plainly. On one RTX 4050 and one 100k×384 workload, shipped CPU and WGPU exhaustive search exchange the lead between 2,000 and 3,000 eligible vectors. The route also depends on how the predicate is represented: compact rows win over shader-side evaluation when the eligible set is tiny, then lose when it is dense. At the dense endpoint, exhaustive WGPU is competitive with one tuned USearch point, but the latency interval crosses parity. Android and Intel Arc repeat the dense CPU/GPU ordering without closing the phase-space gaps. The A6000 study is useful precisely because it disagrees at $E \approx 10k$: FAISS wins there, and an unrelated CUDA formulation wins only later.

The defensible outcome is a bounded routing map. It is more useful than a universal Qenlo victory because a router has to survive the cells where its preferred backend loses.

Reproducibility statement

For the primary native, Linux, and A6000 cohorts, the repository retains raw per-call samples alongside run summaries, configurations, tuning choices, oracle outputs, checksums, environment captures, failed backends, processed tables, plotting code, and reproduction commands. Android evidence is retained as the supplied aggregate records. Generated figures are derived from CSV/JSON artifacts by `research/scripts/analyze_results.py` and `generate_plots.py`. Appendix B maps every headline claim to its artifact family.

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A Complete retained result tables

A.1 Native Windows endpoint and crossover cells

Table 3: Windows 100k×384 endpoint context. Each row contains five runs and 25,000 timed held-out calls; Chroma uses its own storage/API boundary.

Eligible	System	P95 ms	Recall@10	Search
1,000	CPU exhaustive	0.2304	1.00000	exhaustive
1,000	WGPU compact rows	0.6766	1.00000	exhaustive
1,000	WGPU predicate	2.8832	1.00000	exhaustive
100,000	CPU exhaustive	16.6567	0.99998	exhaustive
100,000	WGPU compact rows	4.2428	0.99998	exhaustive
100,000	WGPU mask	4.4591	0.99998	exhaustive
100,000	WGPU predicate	3.2404	0.99998	exhaustive
100,000	USearch, ef=128	3.6245	0.99224	ANN
100,000	Chroma, ef=128	1.1303	0.99414	ANN

Table 4: Post-hoc native crossover sweep. Each row contains five runs and 25,000 held-out calls per backend.

Eligible	CPU P95 ms	WGPU-row P95 ms	CPU/WGPU	Recall@10
2,000	0.5198	0.6340	0.8199	1.0
3,000	0.9563	0.8144	1.1742	1.0
4,000	1.3305	1.0737	1.2392	1.0
6,000	1.9735	1.6033	1.2309	1.0
8,000	2.5003	2.1508	1.1625	1.0
10,000	2.8104	2.6041	1.0792	1.0

A.2 Android device-lab aggregates

Table 5: All 21 Android cells. B is batch size; n is the reported sample count. Every cell reports recall@10=1.0 and no fallback.

Suite	Cell	Rows	B	n	P50	P95	P99 ms
quick	CPU exhaustive	10k	1	16	1.841	5.767	5.767
quick	WGPU exhaustive	10k	1	16	8.707	12.487	12.487
quick	WGPU exhaustive	10k	8	16	5.886	5.886	5.886
quick	IVF-Flat	10k	1	16	3.616	8.056	8.056
quick	IVF-SQ8	10k	1	16	4.746	7.316	7.316
quick	auto selective CPU	10k	1	16	0.023	0.025	0.025
quick	auto selective WGPU	10k	8	16	2.310	2.310	2.310
full	CPU exhaustive	100k	1	64	18.308	29.436	41.559
full	WGPU exhaustive	100k	1	64	13.806	16.975	48.375
full	WGPU exhaustive	100k	8	64	10.608	11.101	11.101
full	IVF-Flat	100k	1	64	8.851	21.617	32.431
full	IVF-SQ8	100k	1	64	20.394	26.578	29.126
full	auto selective CPU	100k	1	64	0.256	0.311	0.323
full	auto selective WGPU	100k	8	64	10.039	15.571	15.571
soak	CPU exhaustive	100k	1	512	25.799	27.019	27.572
soak	WGPU exhaustive	100k	1	512	14.274	17.678	25.769
soak	WGPU exhaustive	100k	8	512	10.290	11.131	12.388
soak	IVF-Flat	100k	1	512	13.977	21.453	24.549
soak	IVF-SQ8	100k	1	512	27.359	37.090	40.581
soak	auto selective CPU	100k	1	512	0.357	0.544	0.743
soak	auto selective WGPU	100k	8	512	8.174	9.890	12.441

The automatic rows use a selective predicate and cannot be compared with dense rows. Their fixed-policy counterfactuals were not retained, so routing regret is not computable. Upload/readback/allocation diagnostics remain in `android_device_lab.csv`.

A.3 A6000 result distributions

Table 6: Real TopK strict-filter cell, $N = 1M$, $D = 768$, $E = 10{,}026$, 5,000 calls/system.

System	P50	P95	P99 ms	Recall@10	Gate
NumPy CPU exhaustive	0.1217	0.1973	0.2283	1.0	pass
FAISS GPU Flat	0.1261	0.1322	0.1458	1.0	pass
research CUDA exhaustive prototype	0.1600	0.1707	0.1809	1.0	pass
cuVS brute force	0.3615	0.3846	0.4131	0.9	fail

Table 7: Held-out synthetic A6000 completed-call distribution, 500 calls/system/cell. QPS is derived from mean latency.

Eligible	System	P50	P95	P99 ms	QPS
1k	FAISS Flat	0.0672	0.0857	0.0952	14,446
1k	research CUDA prototype	0.0775	0.0975	0.1155	12,303
10k	FAISS Flat	0.1272	0.1373	0.1505	7,854
10k	research CUDA prototype	0.1438	0.1557	0.1741	6,911
100k	FAISS Flat	0.7755	0.7887	0.8001	1,286
100k	research CUDA prototype	0.5559	0.5673	0.5763	1,797
1M	FAISS Flat	6.2736	6.3475	6.4357	159
1M	research CUDA prototype	4.6508	4.6683	4.7364	214

A.4 Descriptive Linux library cohort

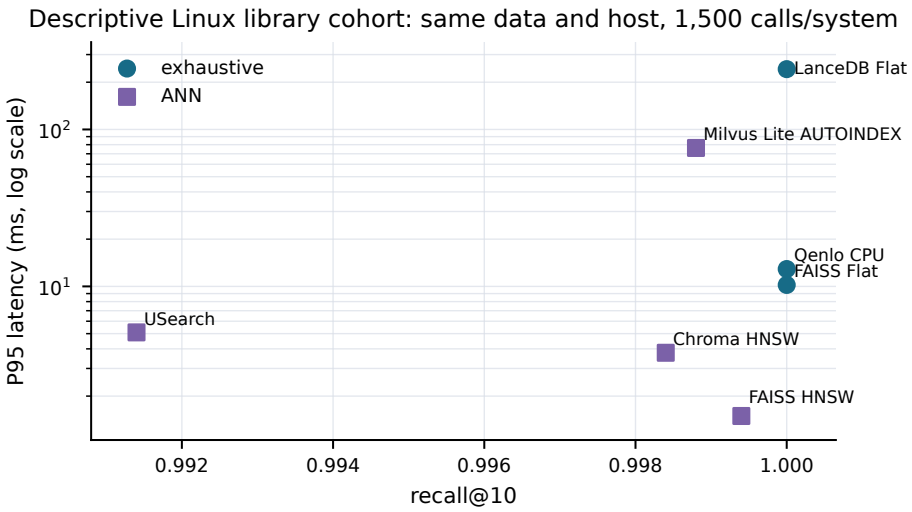

Figure 5: Descriptive Linux cohort. Latency and recall are comparable within this host, but service and persistence semantics differ.

Table 8: All-row Linux 100k×384 cohort, three runs and 1,500 calls/system.

System	Search	P95 ms	Recall@10
FAISS HNSW	ANN	1.4939	0.9994
Chroma HNSW	ANN	3.7767	0.9984
USearch	ANN	5.0933	0.9914
FAISS Flat	exhaustive	10.2171	1.0000
Qenlo CPU	exhaustive	12.9014	1.0000
Milvus Lite AUTOINDEX	ANN	76.2294	0.9988
LanceDB Flat	exhaustive	242.2557	1.0000

B Artifact map

Table 9: Claim-to-artifact map. Paths are relative to the repository root.

Claim family	Authoritative artifact
Native Windows endpoints	`benchmarks/2026-08-28/real/*/samples.csv`, `runs.csv`, `summary.*`
Native 2k–10k crossover	`research/data/raw/2026-09-02-native-crossover/`
Android aggregates	Processed CSV under `research/data/processed/`; raw provenance note under `research/data/raw/2026-09-02-android/`
Linux libraries	`benchmark-results/2026-09-02-runpod-archive/retained-archive/artifacts/`
Real A6000 strict gate	`benchmark-results/2026-09-02-runpod-a6000-strict-research-gate/`
Synthetic A6000 sweep	`research/data/raw/2026-09-02-a6000-cuda/heldout/`
Processed tables	`research/data/processed/`
Analysis and figures	`research/scripts/analyze_results.py`, `generate_plots.py`
Failures and exclusions	`research/methodology-audit.md`, `completion-audit.md`

C Minimum next experiment

The missing experiment is a same-machine phase diagram with fixed data and query order. It should test eligible counts of 500, 1k, 2k, 3k, 5k, 10k, 25k, 50k, and 100k; dimensions 384 and 768; and batches 1, 8, and 32. Every cell should compare optimized CPU exhaustive, shipped WGPU exhaustive, FAISS GPU Flat, a tuned HNSW recall–latency curve, and at least one filter-aware ANN method. The six headline cells—CPU and WGPU at 2k and 3k, dense WGPU, and dense USearch—should use at least 20 independent complete runs. A compact-row ablation should compare the tested per-call materialization path with a cached or resident eligible list, while retaining identical scoring and completed-call boundaries. Matched fixed-policy runs would make routing regret measurable. Repeating identical E at different N would test H2; correlated and anti-correlated metadata would test whether selectivity alone predicts ANN difficulty.